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Both authors performed the following: conception and design of the study, analyzing and interpreting data, drafting and revising the manuscript, and approval of the final version of the manuscript.

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CASE REPORT

Rehabilitation Nutrition for Possible Sarcopenic Dysphagia After Lung Cancer Surgery

A Case Report

ABSTRACT

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Sarcopenic dysphagia is characterized by the loss of swallowing muscle mass and function associated with generalized loss of skeletal muscle mass and function. In this report, the authors describe a patient with possible sarcopenic dysphagia after lung cancer surgery and was treated subsequently by rehabilitation nutrition. A 71-year-old man with lung cancer experienced complications of an acute myocardial infarction and pneumonia after surgery. He was ventilated artificially, and a tracheotomy was performed. The patient received diagnoses of malnutrition, severe sarcopenia, and possible sarcopenic dysphagia. His dysphagia was improved by a combination of dysphagia rehabilitation including physical and speech therapy and an improvement in nutrition initiated by a nutrition support team. Finally, he no longer had dysphagia and malnutrition. Sarcopenic dysphagia should be considered in patients with sarcopenia and dysphagia. Rehabilitation nutrition using a combination of both rehabilitation and nutritional care management is presumptively useful for treating sarcopenic dysphagia.

Key Words: Deglutition Disorder, Intensive Care Unit, Malnutrition, Sarcopenia

Sarcopenic dysphagia is caused by a loss of both swallowing and generalized skeletal muscle mass and function.^{1,2} Age-related loss of muscle mass of the intrinsic muscle of the tongue,³ geniohyoid muscle,⁴ and pharyngeal lumen size⁵ has been studied in older people. Mid-upper arm circumference and calf circumference were shown to correlate with dysphagia.^{6,7} Sarcopenia has been reported to be an independent risk factor of dysphagia in hospitalized older patients.⁸ Furthermore, an association between decreased tongue pressure, sarcopenia, and sarcopenic dysphagia has also been observed in these patients.⁹ Therefore, sarcopenic dysphagia may be common in older people with sarcopenia and dysphagia^{1,2} and an important public health issue.

The European Working Group on Sarcopenia in Older People categorized sarcopenia into age-, activity-, disease-, or nutrition-related sarcopenia.¹⁰ Primary sarcopenia is considered to be age related, whereas secondary sarcopenia should

be considered when 1 or more other causes are evident, such as activity-, disease-, or nutrition-related sarcopenia. Patients with sarcopenia and sarcopenic dysphagia can simultaneously experience activity-, disease-, and nutrition-related sarcopenia.

The treatment of sarcopenia and sarcopenic dysphagia depends on its etiology and requires rehabilitation nutrition.^{2,11} Rehabilitation nutrition is a combination of rehabilitation and nutrition care management to maximize functionality in people with disability.² However, to the authors' knowledge, no case of sarcopenic dysphagia treated by rehabilitation nutrition has been reported. In this article, the authors report a case of possible sarcopenic dysphagia after lung cancer surgery.

Case

A 71-year-old man was admitted to the hospital because of lung cancer in the right upper lobe. He had chronic obstructive pulmonary disease and required home oxygen therapy. However, he had not had dysphagia, and his activities of daily living were independent before surgery. He has received a diagnosis of lung cancer 68 days before surgery. A resection of the right-upper lung, chest walls, and lymph nodes was performed, and he was then transferred to the intensive care unit (ICU). Complications including an acute myocardial infarction and pneumonia occurred the day after surgery, requiring intubation and artificial ventilation. Sedative drugs were administered from days 4 to 6 and from days 8 to 10 (Table 1). A tracheotomy was performed on day 9, and he was weaned from mechanical ventilation on day 11 and discharged from the ICU to a hospital ward on day 14. During the ICU period, he developed dysphagia classified as severe with a Functional Oral Intake Scale (FOIS) of level 1 (nil by mouth).¹² Consciousness was clear when sedative drugs were not administered. His C-reactive protein level was between 4 and 14 mg/dL, indicating the presence of moderate to severe inflammation. The patient stayed in his bed at all times during the ICU period.

On day 16, he was referred to the Department of Rehabilitation Medicine, where he was prescribed physical and speech therapy by a physiatrist. His Barthel index score was 27 of 100 points, and mobility on level surfaces score was 0 of 15 points, indicating he was not able to walk independently. Dysphagia was still classified as severe with an FOIS of level 1. On day 21, videoendoscopic evaluation of swallowing showed bilateral vocal cord paralysis and saliva pooling in the pharynx. On day 39, videoendoscopic evaluation of swallowing revealed that the right vocal cord paralysis had resolved. Radiation therapy for lung cancer,

diagnosed by pathological examination as a pleomorphic carcinoma, was performed between days 43 and 88. During this period of radiation therapy, the patient experienced pharyngeal pain and nausea. Chemotherapy was not performed. On day 53, the tracheotomy tube was removed, although his dysphagia remained severe with an FOIS of level 1. A percutaneous endoscopic gastrostomy was therefore performed on day 74. The rehabilitation team predicted that if the patient's nutrition status could not be improved a worse outcome of dysphagia would occur. Therefore, on day 80, he was referred to the nutrition support team that consisted of doctors, nurses, registered dietitians, pharmacists, clinical technologists, physical therapists, occupational therapists, and a speech therapist.

At referral to the nutrition support team, his height was 174.5 cm; body weight, 46.6 kg; and body mass index, 15.3 kg/m². His body weight at admission had been 56.8 kg, equating to a weight loss of 18%. Laboratory tests showed a serum albumin of 3.3 g/dL, hemoglobin of 12.4 g/dL, and C-reactive protein of 0.86 mg/dL, indicating the presence of malnutrition, anemia, and low-grade inflammation, respectively. The Mini Nutritional Assessment Short Form^{13–15} score was 2, indicating malnourishment.

Skeletal muscle mass was assessed by measuring the cross-sectional area of the left and right psoas muscles using abdominal computed tomography (Slice-O-Matic software version 5.0; Tomovision, Magog, Quebec, Canada) at the caudal end of the third lumbar vertebra.^{16,17} Simple psoas cross-sectional area measurement is a quick and easy method for assessing sarcopenia.¹⁸ The muscles were quantified within a Hounsfield unit range of -29 to 150. The skeletal muscle index was defined as total psoas muscle area divided by height squared. On day 16, the patient's left and right psoas muscle areas were 7.249 and 6.797 cm², respectively, and skeletal muscle index was 4.61 cm²/m². The cutoff value for the skeletal muscle index in Japanese cancer patients is defined as less than 5.67 cm²/m² in males and less than 3.95 cm²/m² in females.¹⁹ Therefore, the patient was classified as having a lower muscle mass. At referral to the nutrition support team, his grip strengths were 14.5 kg (right) and 26.3 kg (left), although he was right-handed. His usual gait speed was 0.4 m/s assessed by the 10-m walk test. Using the criteria of the European Working Group on Sarcopenia in Older People¹⁰ and the Asian Working Group for Sarcopenia,²⁰ he received a diagnosis of severe sarcopenia because of his lower levels of muscle mass, muscle strength (grip strength <26 kg in males), and physical function (usual gait speed <0.8 m/s). He also received a diagnosis of possible

sarcopenic dysphagia according to the consensus diagnostic criteria for sarcopenic dysphagia (Table 2),¹ because he had dysphagia and generalized sarcopenia. Despite the presence of left vocal cord paralysis, the main cause of dysphagia was considered to be sarcopenia.

The patient's total energy expenditure was calculated as 1800 kcal by Long and colleagues²¹ formula, basic energy expenditure as 1147 kcal, gross activity factor as 1.3, and gross injury factor as 1.2. However, energy intake at referral to the nutrition support team was 986 kcal, and it was therefore planned to increase energy intake to 1800 kcal through a gastrostomy. Speech therapy that included direct exercises using modified food and indirect exercises such as stretch and resistance training of the swallowing muscles was carried out twice a week during hospitalization. Stretching and resistance training of the swallowing muscles were performed on the days between the speech therapy sessions. Physical therapy that included resistance training, movement exercises, and ambulation exercises was also initiated 5 times a week during hospitalization.

On day 87, his dysphagia improved to FOIS level 3 (tube dependent with consistent oral intake of food or liquid). In contrast, his body weight remained at 46.6 kg, necessitating an increase in energy intake to 2200 kcal to improve malnutrition, sarcopenia, and dysphagia.

The improvement in the patient's dysphagia was associated with improved nutritional intake and an attenuation of the adverse effects of radiation therapy. On day 120, his dysphagia had improved to FOIS level 5 (total oral diet with multiple consistencies, but requiring special preparation or compensations), and therefore the gastrostomy tube was removed. On day 131, he was discharged to

his home with an FOIS level 6 (total oral diet with multiple consistencies without special preparation, but with specific food limitations). At discharge, his body weight was 48.5 kg, serum albumin was 3.4 g/dL, hemoglobin was 12.4 g/dL, C-reactive protein was 0.05 mg/dL, and Barthel index score was 100 points. His grip strengths were 15 kg (right) and 22 kg (left), and usual gait speed was 0.83 m/s, indicating sarcopenia. The Mini Nutritional Assessment Short Form score was 7, consistent with malnourishment. There were no adverse or unanticipated events after rehabilitation and interventions by the nutrition support team.

After discharge, the patient was prescribed oral nutritional supplements and outpatient rehabilitation including physical and speech therapy once or twice a month for 4 months. On day 242, an abdominal computed tomography was performed that showed the left and right psoas muscle areas were 7.524 and 7.361 cm², respectively. The skeletal muscle index was calculated as 4.89 cm²/m², indicating a lower muscle mass. On day 253, his body weight was 55.5 kg, serum albumin was 4.5 g/dL, hemoglobin was 14.5 g/dL, C-reactive protein was 0.09 mg/dL, and grip strengths were 20 kg (right) and 28 kg (left). The Mini Nutritional Assessment Short Form score was 11, indicating a risk of malnutrition, whereas the FOIS was level 7 (total oral diet with no restrictions). However, his left vocal cord paralysis remained. Physical therapy and speech therapy were stopped because he no longer had malnutrition and dysphagia. Despite the patient's muscle mass and strength having improved substantially, he was still classified as having sarcopenia because of the continuing low levels of these 2 parameters.

TABLE 1 Timeline of the patient's clinical course and treatment

Day	0	1	4	6	8	10	14	16
Clinical findings	Surgery	Myocardial infarction, pneumonia, intubation	Sedative drugs started	Sedative drugs stopped	Sedative drugs started	Sedative drugs stopped	Transferred to hospital ward	Bedside physical and speech therapy started
Location	ICU	ICU	ICU	ICU	ICU	ICU		Hospital ward
FOIS ¹²	1	1	1	1	1	1	1	1
BW, kg		56.8					54.1	
BMI, kg/m ²		18.7					17.7	
BI								27
HS, kg								
GS, m/s								0
MNA-SF	5	5					2	

BI, Barthel index; BMI, body mass index; BW, body weight; GS, gait speed; HS, handgrip strength; MNA-SF, Mini Nutritional Assessment Short Form.^{13–15}

DISCUSSION

This article reveals 2 important clinical issues. First, sarcopenic dysphagia may present after lung cancer surgery, and second, rehabilitation nutrition is presumptively useful for treating this condition.

The patient received a diagnosis of possible sarcopenic dysphagia. Major causes of dysphagia are considered to be sarcopenia and oropharyngeal and laryngeal trauma caused by endotracheal and tracheostomy tubes. The patient still had left vocal cord paralysis at the end of outpatient rehabilitation, although his dysphagia had lessened with improvements in malnutrition and sarcopenia. This result indicated that the most likely cause of his dysphagia was sarcopenic dysphagia despite him not being diagnosed definitely with this condition (Table 2). Skeletal muscle mass assessed by psoas muscle area using abdominal computed tomography has been shown to be associated with severe dysphagia in cancer patients.²² It is therefore important that sarcopenic dysphagia is considered in patients with dysphagia and sarcopenia.

Exacerbation in activity-, disease-, and nutrition-related sarcopenia of generalized skeletal muscles and swallowing muscles may develop into sarcopenic dysphagia. In contrast, primary sarcopenia does not exacerbate rapidly, because the cause of primary sarcopenia is aging. Activity-related sarcopenia, disuse muscle atrophy, and muscle weakness may develop in our patient during his stay in the ICU, because his activities of daily living were independent before surgery. It has been reported that early commencement of oral intake is associated with total oral intake at discharge in hospitalized older individuals with pneumonia.²³ Early rehabilitation also can prevent activity-related sarcopenia. Disease-related sarcopenia in our

TABLE 2 Consensus diagnostic criteria for sarcopenic dysphagia¹

- (1) Presence of dysphagia
 - (2) Presence of generalized sarcopenia (generalized loss of skeletal muscle mass and strength)
 - (3) Results of imaging tests (computed tomography, magnetic resonance imaging, ultrasonography) consistent with a loss of swallowing muscle mass
 - (4) The causes of dysphagia without sarcopenia are excluded
 - (5) The cause of dysphagia is considered to be sarcopenic dysphagia, despite other causes of the condition existing, such as stroke, brain injury, neuromuscular diseases, head and neck cancer, or connective tissue diseases
- Definite diagnosis: 1, 2, 3, 4
 Probable diagnosis: 1, 2, 4
 Possible diagnosis: 1, 2, 5

patient was exacerbated by lung surgery, acute myocardial infarction, and pneumonia. Lung cancer and chronic obstructive pulmonary disease can also be causes of disease-related sarcopenia.¹⁰ Nutrition-related sarcopenia in the patient also worsened during treatment, because his energy intake prior to referral to the nutrition support team did not satisfy basic energy expenditure. Adequate caloric and protein intake after surgery is important for optimal recovery, because inadequate caloric and protein intake can lead to complications and a suspension in diet.²⁴

Rehabilitation nutrition may be useful for treating sarcopenic dysphagia. Because the etiology of sarcopenia is multifactorial, collaboration between the rehabilitation and nutrition support teams is useful for assessment and treatment of sarcopenia. In addition, the association between

39	42	43	80	87	88	120	131	253
Right vocal cord paralysis resolved	Gymnasium physical and speech therapy started	Radiotherapy started	Nutritional support team consulted	Increase in energy, protein intake	Radiotherapy stopped	Gastrostomy tube removed	Discharge home	Physical and speech therapy stopped
Hospital ward 1	Hospital ward 1	Hospital ward 1	Hospital ward 1	Hospital ward 3	Hospital ward 3	Hospital ward 5		Outpatient 7
49.7	49.4	49.6	46.6	46.6		48.6	48.5	55.5
16.3	16.2	16.3	15.3	15.3		15.9	15.9	18.2
	42						100	100
	14.5/26.3							20/28
	0.4						0.8	
2	3					7	7	11

malnutrition, sarcopenia, and dysphagia^{25,26} makes rehabilitation nutrition indispensable in the treatment of sarcopenic dysphagia.

In conclusion, sarcopenic dysphagia may present after lung cancer surgery. The case showed that sarcopenic dysphagia must be considered in patients with dysphagia and sarcopenia and that rehabilitation nutrition may be useful for treating this condition. Further investigations are necessary to determine whether sarcopenic dysphagia is a frequent occurrence in descriptive studies and whether rehabilitation nutrition contributes to the treatment of the disorder in intervention studies.

The patient provided informed consent to publication of his case. The investigations and case study were performed in accordance with the ethical standards established in the 1964 Declaration of Helsinki and later amendments and ethical guidelines for medical and health research involving human subjects in Japan.

Supplementary Checklist

CARE Checklist: <http://links.lww.com/PHM/A195>

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